Type MS · Variables, constants, arrays and records

Answer 1 of 9 · Question 1 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c.ii in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

1 of 23

Type MS · Variables, constants, arrays and records

Answer 1 of 9 · Question 1 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
1  # Q01bii
2
3  # initialise counters
4  ballCount = 0
5  greenCount = 0
6  redCount = 0
7
8  ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
9              "green", "green", "green", "red", "green", "red", "green", "red",
10             "red", "green", "green", "red", "red", "green", "green", "red",
11             "green", "red", "red", "green", "green", "red", "green", "green",
12             "red", "green", "green", "green", "red", "green", "red","green"]
13
14 # iterate over ball array
15 for ball in ballArray:
16     ballCount = ballCount + 1
17     if ball == "green":
18         greenCount = greenCount +1
19     else:
20         redCount = redCount + 1
21
22 print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
23       "balls \n", "Total: ", ballCount, "balls")
```

2 of 23

Type MS · Variables, constants, arrays and records

Answer 1 of 9 · Question 1 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

3 of 23

Type MS · Variables, constants, arrays and records

Answer 1 of 9 · Question 1 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

4 of 23

Type MS · Variables, constants, arrays and records

Answer 1 of 9 · Question 1 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part d in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part d in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part d in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part d in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part d in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 8		(1)

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part d in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part d in this mark scheme extract

1 (e) (iv)	E4	message		(1)

4 of 16

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part d in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part d in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
	C3	2.1.5	Correction to print statement (1)		
Code example					
Python					

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

9 of 16

Type MS · Variables, constants, arrays and records

Answer 2 of 9 · Question 1 part d · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part d in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 1 of 10 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (a)	A1	**1. The only correct answer is B** ***A*** *is not correct because as it is an arithmetic operator* ***C*** *is not correct because as it is a relational operator* ***D*** *is not correct because as it is a relational operator*		(1)

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 2 of 10 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2	Award up to 2 marks for a linked description such as: • 1D represents items as a list (1), 2D as a table / 2D as rows and columns (1) • 1D is a row/column (1), 2D is a table / 2D has rows and columns (1) • Each element in 1D is a single value (1), each element in 2D is a 1D array (1) • 1D array can only store one type of element (1). 2D array can store multiple elements in it (1)	Accept an interpretation	(2)

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 3 of 10 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (c)	C1	Change num_twenties == to num_twenties = (1)		
	C2	The **left over** variable named the same in both places (1)		
	C3	Add + before str(left_over) over in final print statement (1)	Accept equivalent e.g. ,	(3)

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 4 of 10 · find part b in this mark scheme extract

Question	Typ	Answer	Additional Guidance	Mark
1 (d) (i)	D1	Award 1 mark for adding an appropriate comment at the end of the line where there is relational operator: `if vowel == letter:  # relational operator and selection`	May be on different line numbers The comment(s)	(1)

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 5 of 10 · find part b in this mark scheme extract

1 (d) (ii)	D2	Award one mark for adding an appropriate comment at the end of a line where iteration starts:
		`for letter in sentence: # iteration starts`
		`for vowel in vowels: # iteration starts`
			added must clearly identify the component

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 6 of 10 · find part b in this mark scheme extract

1 (d) (iii)	D3	Award one mark for adding an appropriate comment at the end of the line where selection starts: `if vowel == letter:  # relational operator and selection`	

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 7 of 10 · find part b in this mark scheme extract

1 (d) (iv)	D4	Award one mark for adding an appropriate comment at the end of a line where a data structure is initialised:
		`vowels = ["a","e","i","o","u"] # data structure initalised`
		`numVowels = [0,0,0,0,0] # data structure initalised`
				(1)

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 8 of 10 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 9 of 10 · find part b in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td rowspan="6">2 (b)</td><td colspan="2">Award 1 mark for each correct condition.</td><td rowspan="6">Accept alternatives e.g. Line 11 booksSold <=4 etc.</td><td rowspan="6">(4)</td></tr>
<tr><td></td><td>Condition</td><td>Output message</td></tr>
<tr><td>B1</td><td>Number of books sold is under 5 or profit made is under 5</td><td>Poor performance this week</td></tr>
<tr><td>B2</td><td>Number of books sold is over 20; profit made is at least 20</td><td>Sales and profit are excellent this week</td></tr>
<tr><td>B3</td><td>Number of books sold is at least 5; profit made is at least 10</td><td>Sales and profit are good this week</td></tr>
<tr><td>B4</td><td>All other inputs</td><td>Alert manager</td></tr>
<tr><td colspan="5">Code example</td></tr>
<tr><td>Python</td><td colspan="4">

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

</td></tr>
</table>

Type MS · Variables, constants, arrays and records

Answer 3 of 9 · Question 1 part b · 2 marks

2021 JUN · Paper 02 · extract 10 of 10 · find part b in this mark scheme extract

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 9 · find part a in this mark scheme extract

1 (a)	A1	**The only correct answer is D** A is not correct because the value can be used more than once B is not correct because the value does not always have to be input C is not correct because the value does not always have to be used in a calculation		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 2 of 9 · find part a in this mark scheme extract

1 (b)	B1	**The only correct answer is B** A is not correct because this would not aid readability of the code C is not correct because this would not aid readability of the code D is not correct because this would not aid readability of the code		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 3 of 9 · find part a in this mark scheme extract

<table>
<tr><td>1 (c)</td><td>C1
C2</td><td>Award 1 mark for each of:
<table><tr><th>Data type</th><th>Example value</th></tr><tr><td>char</td><td>An example of any single letter, number or symbol</td></tr><tr><td>real</td><td>An example of a number with a decimal point</td></tr></table></td><td>Accept quotes around the character

Real number does not need a number after the decimal point to be awarded the mark</td><td>(2)</td></tr>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
</table>

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 4 of 9 · find part a in this mark scheme extract

1 (d)	D1	**Award up to 2 marks for a description such as:** • Boundary testing uses the most extreme valid data / minimum and maximum valid values (1) whereas erroneous testing uses invalid data (1)		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 5 of 9 · find part a in this mark scheme extract

1 (e)	E1	Logic/logical		(1)

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 6 of 9 · find part a in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 7 of 9 · find part a in this mark scheme extract

1 (f)	**Award 1 mark for each of:**			
	F1	Bracket added and no other amendments made to the line (1)		
	F2	== becomes = (1)		
	F3	" moved to after **is** and before , OR , area changed to +str(area) (1) Must still include the print statement		(3)
Code examples				
Python	3 length = float(input("Enter the first number ") 4 5 area = length * length 6 7 print("The area of the square is" , area)			

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 8 of 9 · find part a in this mark scheme extract

1 (g)	**Award 1 mark for each of:**			
	G1	At least 1 condition correct (1)		
	G2	At least 1 > or < condition and message match (1)		
	G3	At least 1 output includes letter and stored letter with an appropriate message (1)		
	G4	All conditions and outputs correct (1)		(4)
Code examples				
Python				

```
# Amend the code by completing the if statement
if letter > storedLetter:
    print(letter + " is later in the alphabet than " + storedLetter)
elif letter == storedLetter:
    print(letter + " is the same as " + storedLetter)
else:
    print(letter + " is earlier in the alphabet than " + storedLetter)
```

Type MS · Variables, constants, arrays and records

Answer 4 of 9 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 9 of 9 · find part a in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 5 of 9 · Question 1 part c.i · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part c.i in this mark scheme extract

<table>
<tr>
<td>1(a)</td>
<td>A1
A2
A3</td>
<td>Award one mark for each correct cell:

<table>
<tr><td>Input</td><td>Debit / Credit / Description</td></tr>
<tr><td>Processing</td><td>Balance + Credit / Balance - Debit</td></tr>
<tr><td>Output</td><td>Balance</td></tr>
</table>
</td>
<td>Input: accept examples of inputs e.g. 10.23 (must be in Figure 1)

Processing; mark first 3 things (ignore extra text) if correct e.g.
balance + credit – debit (1)
balance + credit + debit (1)
balance – credit (0)
accept examples from table e.g. 128.35 – 10.23 (1)

Output: accept description of formula e.g. subtract debit from balance (1)</td>
<td>3</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Variables, constants, arrays and records

Answer 5 of 9 · Question 1 part c.i · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Variables, constants, arrays and records

Answer 5 of 9 · Question 1 part c.i · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 6 of 9 · Question 1 part c.ii · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part c.ii in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Variables, constants, arrays and records

Answer 6 of 9 · Question 1 part c.ii · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Variables, constants, arrays and records

Answer 6 of 9 · Question 1 part c.ii · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 1 of 16 · find part d in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 2 of 16 · find part d in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 3 of 16 · find part d in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	**4**

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 4 of 16 · find part d in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 5 of 16 · find part d in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 6 of 16 · find part d in this mark scheme extract

Python

```
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 7 of 16 · find part d in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 8 of 16 · find part d in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 9 of 16 · find part d in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 10 of 16 · find part d in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 11 of 16 · find part d in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 12 of 16 · find part d in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 13 of 16 · find part d in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 14 of 16 · find part d in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 15 of 16 · find part d in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 7 of 9 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · extract 16 of 16 · find part d in this mark scheme extract

<table>
<tr>
<td>1(e)</td>
<td>E1
E2</td>
<td>One mark for each correct cell:
<table>
<tr><th>Item</th><th>Data type</th></tr>
<tr><td>45.82</td><td>All programming languages:
real/float
C#:
decimal/double
Java:
Double/double/BigDecimal
Python:
decimal</td></tr>
<tr><td>True</td><td>All programming languages:
Boolean/bool</td></tr>
</table></td>
<td>Ignore capitalisation
Do not accept string for True</td>
<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 1 of 14 · find part a.ii in this mark scheme extract

2(a)(i)	A1	**The only correct answer is A** ***B*** *is not correct because* ***computational thinking*** *are skills used in creating algorithms and programs* ***C*** *is not correct because* ***decomposition*** *is breaking down a problem into smaller parts* ***D*** *is not correct because* ***pseudocode*** *is a notation for expressing an algorithm*		**1**

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 2 of 14 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 3 of 14 · find part a.ii in this mark scheme extract

2(a)(ii)	A2	Award **one** mark for: • Variable (1)	Allow parameter Ignore local/global	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 4 of 14 · find part a.ii in this mark scheme extract

2(b)(i)	B1	Award **one** mark for any of the following: • An instruction is impossible to execute (1) • The program stops executing **unexpectedly** / **crashes** (1)	Do not award an example, as that is the next question Don't award repeat of question e.g. "Stopping during execution" without clarifying reason or that stop was unexpected	**1**

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 5 of 14 · find part a.ii in this mark scheme extract

2(b)(ii)	B2	Award **one** mark for any of the following: • Division by zero (1) • Accessing memory outside the range of an array (1) • Attempting an operation on incompatible types (1) • File not found (1) • An exception is raised that cannot be/is not handled (1)	Award other examples, if they can be mapped to a bullet	**1**

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 6 of 14 · find part a.ii in this mark scheme extract

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 7 of 14 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 8 of 14 · find part a.ii in this mark scheme extract

2(c)		Award **one** mark for each of:		
	C1	Add or remove a semicolon on initialisation of the length variable (1)	• `int length = 4;` • `int length = 4;` • `length = 4`	
	C2	Add a comma between the arguments to the subprogram call to generate a random number (1)	• `width = randGenerator.Next (1, 6);` • `width = randGenerator.nextInt(1, 6);` • `width = random.randint (1, 5)`	
	C3	Replace modulus operator (%) with multiplication operator (*) (1)	• `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length);` • `perimeter = (2 * width) + (2 * length)`	**3**

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 9 of 14 · find part a.ii in this mark scheme extract

C#

```
// Q02cFINISHED

int width = 0;
int length = 4;
int perimeter = 0;
Random randGenerator = new Random();

width = randGenerator.Next (1, 6);
perimeter = (2 * width) + (2 * length);

Console.WriteLine (width.ToString() + " " +
                   length.ToString() + " " +
                   perimeter.ToString());
```

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 10 of 14 · find part a.ii in this mark scheme extract

Java

```java
// Q02cFINISHED

import java.util.Random;

public class Q02cFINISHED
{
    public static void main (String[] args)
    {
        int width = 0;
        int length = 4;
        int perimeter = 0;
        Random randGenerator = new Random();

        width = randGenerator.nextInt(1, 6);
        perimeter = (2 * width) + (2 * length);

        System.out.println (String.valueOf(width) + " " +
                            String.valueOf(length) + " " +
                            String.valueOf(perimeter));
    }
}
```

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 11 of 14 · find part a.ii in this mark scheme extract

Python

```
# Q02cFINISHED

import random

width = 0
length = 4
perimeter = 0

width = random.randint (1, 5)
perimeter = (2 * width) + (2 * length)

print (width, length, perimeter)

```

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 12 of 14 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 13 of 14 · find part a.ii in this mark scheme extract

<table>
<tr>
<td>2(d)</td>
<td>D1
D2
D3
D4</td>
<td>
<table>
<tr><td>Erroneous</td><td>Award one mark for any of the following up to a maximum of two marks:
Contains incorrect data type (1)
<2 letters (1)
9+ letters (1)</td></tr>
<tr><td>Normal</td><td>A string between 2 and 8 in length that includes only letters (1)</td></tr>
<tr><td>Boundary</td><td>Award one mark for any of the following:
A string of exactly 8 letters (1)
A string of exactly 2 letters (1)</td></tr>
</table>
</td>
<td>Award any correct example (not description)
Do not award an empty cell as equivalent to empty string
Do not accept same test twice for erroneous</td>
<td>4</td>
</tr>
</table>

Type MS · Variables, constants, arrays and records

Answer 8 of 9 · Question 2 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 14 of 14 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Variables, constants, arrays and records

Answer 9 of 9 · Question 1 part c.i · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part c.i in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

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Type MS · Variables, constants, arrays and records

Answer 9 of 9 · Question 1 part c.i · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
1  # Q01bii
2
3  # initialise counters
4  ballCount = 0
5  greenCount = 0
6  redCount = 0
7
8  ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
9              "green", "green", "green", "red", "green", "red", "green", "red",
10             "red", "green", "green", "red", "red", "green", "green", "red",
11             "green", "red", "red", "green", "green", "red", "green", "green",
12             "red", "green", "green", "green", "red", "green", "red","green"]
13
14 # iterate over ball array
15 for ball in ballArray:
16     ballCount = ballCount + 1
17     if ball == "green":
18         greenCount = greenCount +1
19     else:
20         redCount = redCount + 1
21
22 print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
23       "balls \n", "Total: ", ballCount, "balls")
```

2 of 23

Type MS · Variables, constants, arrays and records

Answer 9 of 9 · Question 1 part c.i · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

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Type MS · Variables, constants, arrays and records

Answer 9 of 9 · Question 1 part c.i · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

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Type MS · Variables, constants, arrays and records

Answer 9 of 9 · Question 1 part c.i · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part c.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark